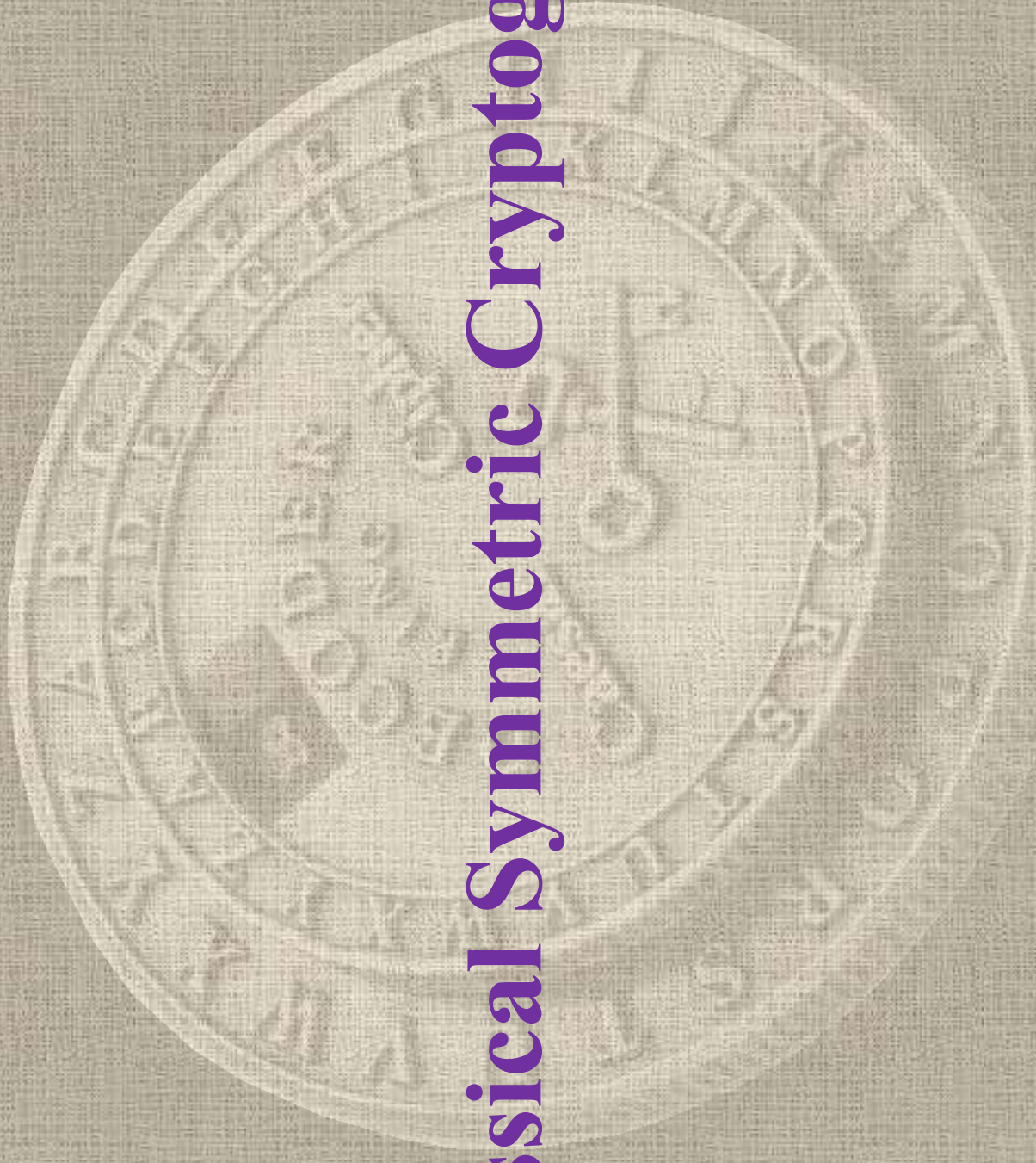


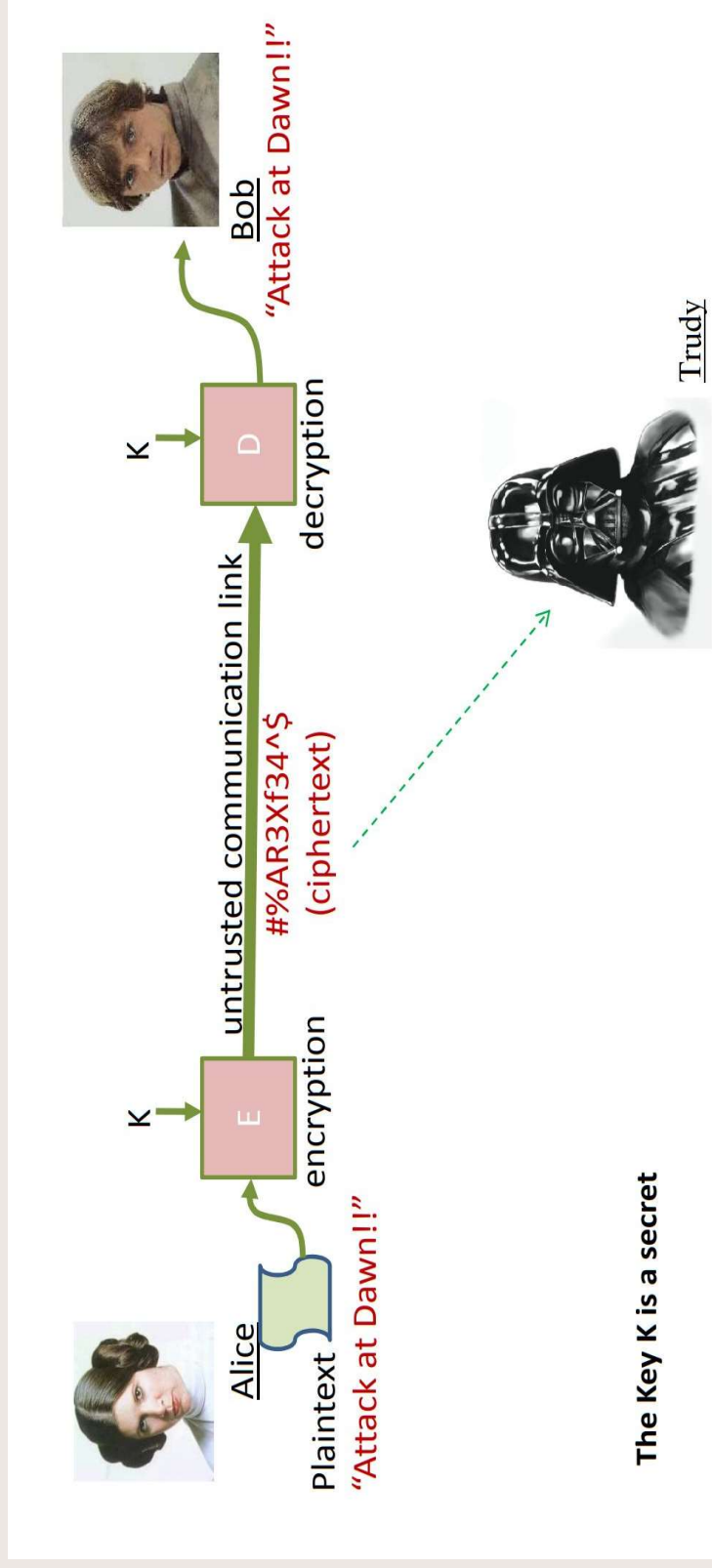
Classical Symmetric Cryptography



Classic Symmetric Cryptography

- Used historically but now has fallen.
- Classical ciphers can be practically computed and solved by hand.
- They are also usually very simple to break with modern technology.
- The classic crypto can be classified into substitution and transposition ciphers.
- The substitution crypto is the oldest cryptosystem that has been used dating back at least 2,000 years.

Symmetric Ciphers



- Uses the same keys for both encryption of plaintext and decryption of cipher text.
- Trudy listen only cipher text and can not get plaintext because he does not know the secret key K.

Cryptosystem

A **cryptosystem** is a five-tuple $(\mathbb{P}, \mathbb{C}, \mathbb{K}, \mathbb{E}, \mathbb{D})$, where the following are satisfied

- $\mathbb{P}$ is a finite set of possible **plaintexts**
- $\mathbb{C}$ is a finite set of possible **ciphertexts**
- $\mathbb{K}$, the key space, is a finite set of possible keys
- $\mathbb{E}$ is a finite set of **encryption functions**
- $\mathbb{D}$ is a finite set of **decryption functions**
- $\forall K \in \mathbb{K}$

Encryption Rule : $\exists e_K \in \mathbb{E}$, and

Decryption Rule : $\exists d_K \in \mathbb{D}$

such that $(e_K: \mathbb{P} \rightarrow \mathbb{C}), (d_K: \mathbb{C} \rightarrow \mathbb{P})$ and $\forall x \in \mathbb{P}, d_K(e_K(x)) = x$

Properties of e_K

- Encryption is **injective** (one-to-one)
- Facts about e_K
- – i.e. $e_k(x_1) = e_k(x_2)$ iff $x_1 = x_2$
- – Why?
- If not, then Bob does not know if the ciphertext came
- from x_1 or x_2
- If $\mathbb{P} = \mathbb{C}$, then the encryption function is a **permutation**
- $\mathbb{C}$ is a rearrangement of $\mathbb{P}$

Kerckhoff's Principle

- The inner working of the cryptosystem are completely known to the attacker.
- The only secret is a key.
- **Why do we make this assumption?**
- – Algorithms can be leaked (secrets never remain secret)
- – or reverse engineered

Classes of Cipher Algorithms

- **Monoalphabetic substitution ciphers** - each letter is replaced with another. Examples include simple substitution, caesar, affine, etc.
- **Polyalphabetic ciphers** - different alphabets are used to encipher letters depending on position. Examples include Porta cipher, Vigenere, Gronsfeld, etc.
- **Polygraphic Substitution ciphers** - groups of characters are replaced. Examples include cipher, playfair, foursquare etc.
- **Transposition ciphers** - these involve permuting only the positions of the characters, leaving the identity of the characters unchanged. examples include Columnar Transposition, double Transposition, route ciphers etc.
- **Other types** - these ciphers may include elements from several of the above classes. Examples include one time pad, codebook etc.

Simple Substitution (Shift) Cipher

- The message is encrypted by substituting the letter of alphabet n places ahead of the letter. To decrypt the ciphertext, simple shift each ciphertext letter backward by n .
- Translate letter of alphabet to numbers,

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Plaintext Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Shift cipher can be described mathematically as.

- Plaintext set : $\mathbb{P} = \{0, 1, 2, 3 \dots, 25\}$
- Ciphertext set : $\mathbb{C} = \{0, 1, 2, 3 \dots, 25\}$
- Key space : $\mathbb{K} = \{0, 1, 2, 3 \dots, 25\}$
- Encryption Rule : $e_K(x) = (x + K) \bmod 26$,
- Decryption Rule : $d_K(x) = (x - K) \bmod 26$

where $K \in \mathbb{K}$ and $x \in \mathbb{P}$

Note: – Each K results in a unique mapping ($e_K: \mathbb{P} \rightarrow \mathbb{C}$), ($d_K: \mathbb{C} \rightarrow \mathbb{P}$) and $d_K(e_K(x)) = x$,

The encryption/decryption rules are permutations

Example, with $n=3$

- The substitution which acts as the key is

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Ciphertext Alphabet	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z

- Plaintext: attackatdawn Ciphertext : DWWDFNDWGDZQ
- The substitution of a shift of three is known as the Caesar's cipher because it was used success by Julius Caesar

Examples

- Text : neverdisclosethesecretkey
- Shift: 15

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w
Ciphertext Alphabet	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L

- Cipher: CTKTG SXHRADHTIWTHTRTGTIZTN

- Text : ATTACKATONCE
- Shift: 20
- Cipher: UNNUWEUNIHWY

How good is the Substitution cipher?

- A good cipher has two properties
 - Easy to compute
 - Satisfied –
 - An attacker (Trudy), who views the ciphertext should not get a information about the plaintext.
 - Not Satisfied!!
 - The attacker needs at-most 26 guesses to determine the secret key
 -
 - This is an exhaustive key search (known as brute force attack)

Cryptanalysis

Trudy's interest is how to get the secret key using some information.

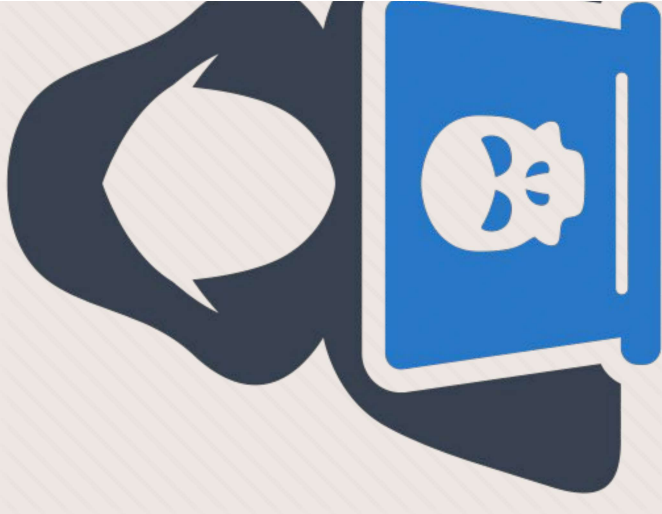
- **Attack models**

- **attack using ciphertext only**

Trudy knows about decryption process, choose the ciphertext and get the plaintext.

- **attack using plaintext only**

Trudy knows about encryption process, choose the plaintext and get the ciphertext.



Cryptanalysis of Simple substitution cipher

- Let ciphertext is EXXEGOEXSRGI
- Using Brute Force
- Try for 26 possible keys.
 - Key =1, DWWDFNDWRQFH
 - Key =2, CVVCEMCVQPEG
 - Key =3, BUUBDLBUPODF
 - Key=4 , **ATTACKATONCE**
 - Key=5, ...
 - Key=25....

Only Key=4, text is meaningful.

History & Usage

- • Used by Julius Caesar in 55 AD with $K=3$. This variant is known as Caesar's cipher.
- • Augustus Caesar used a variant with $K=-1$ and no modular operation.
- Shift ciphers are extremely simple, still used in Modern times
 - – By Russian Soldiers in first world war
 - – Last known use in 2011 (by militant groups)

How large of a key space is large enough

- Trudy has a fast computer that's able to test 2^{40} keys each second.
- key space size 2^{56} ----- 2^{16} seconds, or about 18 hours.
- key space size 2^{64} ----- more than half a year
- key space size 2^{128} would require more than nine quintillion years.
- For modern symmetric ciphers, the key is typically 128 bits or more, giving a key space of size

Substitution Cipher

Substitution cipher can be described mathematically

- Plaintext set : $\mathbb{P} = \{a, b, c, d \dots, z\}$
- Ciphertext set : $\mathbb{C} = \{A, B, C, D \dots, Z\}$
- Key space : $\mathbb{K} = \{\pi \mid \text{such that } \pi \text{ is a permutation of the alphabets}\}$
- Encryption Rule : $e_{\pi}(x) = \pi(x)$,
- Decryption Rule : $d_{\pi}(x) = \pi^{-1}(x)$

Example of Substitution Cipher

Plaintext

Ciphertext

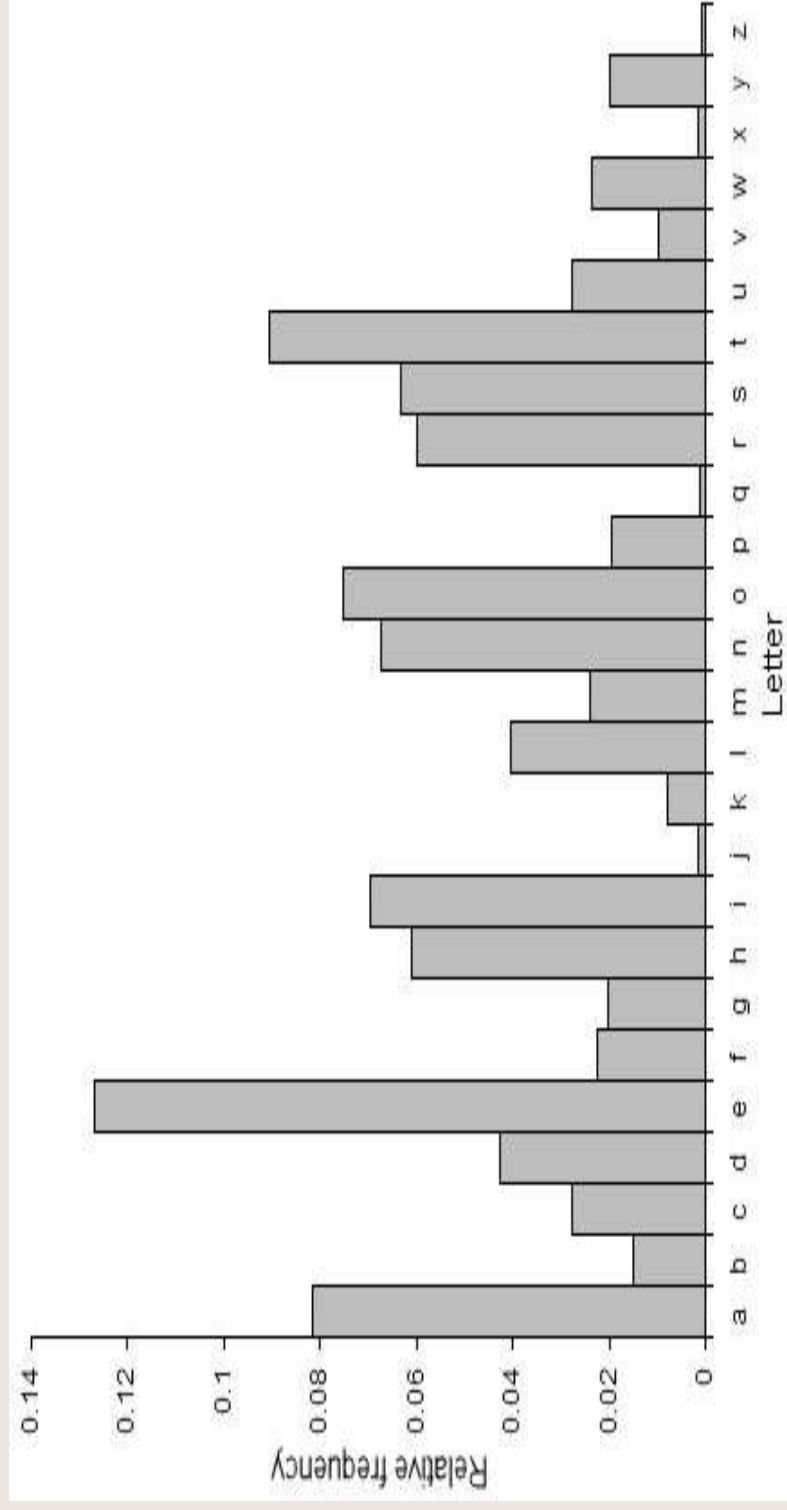
a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
S	R	T	B	Q	C	A	U	P	V	J	X	W	D	Y	I	Z	E	H	J	F	N	L	G	M	O

Plaintext: attackatdawn

Ciphertext: SJJTSJBBSLD

Possible permutations are $26!$. Possible keys are $4 \times 10^{26} \approx 2^{88}$ keys.

Cryptanalysis of a Substitution Using Frequency Analysis



English Letter Frequencies distributed of English language
Frequency based analysis attributed to Al-kindī, an Arab mathematician (in AD 800)

Letter	Frequency
e	0.125
t	0.08
a	0.07
o	0.065
i	0.06
n	0.055
s	0.05
h	0.045
r	0.04
d	0.035
l	0.03
c	0.025
u	0.02
m	0.015
w	0.01
f	0.005
g	0.005
y	0.005
p	0.005
b	0.005
v	0.005
k	0.005
j	0.005
x	0.005
q	0.005
z	0.005

Most Common words known to Ciphe

One-Letter Words	a, I.
Frequent Two-Letter Words	of, to, in, it, is, be, as, at, so, we, he, by, or, on, do, if, n my, up, an, go, no, us, am
Frequent Three-Letter Words	the, and, for, are, but, not, you, all, any, can, had, her, v one, our, out, day, get, has, him, his, how, man, new, n old, see, two, way, who, boy, did, its, let, put, say, she, t use
Frequent Four-Letter Words	that, with, have, this, will, your, from, they, know, want, been, good, much, some, time

Combination of Letters appear in cipher

Most Frequent Single Letters	ETAOINSHRDLU
Most Frequent Digraphs	th er on an re he in ed nd ha at en es of or nt ea ti to it st io le is ou de rt ve
Most Frequent Trigraphs	the and tha ent ion tio for nde has nce edt tis oft sth men
Most Common Doubles	ss ee tt ff ll mm oo
Most Frequent Initial Letters	TOAWBCDSFMRHIYEGLNPUJK
Most Frequent Final Letters	ESTDNRYFLOGHAKMPUW

Cryptanalysis of Simple Substitution Ciphers Using Chi-squared Statistic

- The Chi-squared Statistic is a measure of how similar two categorical probability distributions are.
- If the two distributions are identical, the chi-squared statistic is 0, if the distributions are different, some higher number will result.

- The fitness value of A using chi-squared statistic is

$$\chi^2(C, E) = \sum_{i=A}^{i=Z} \frac{(C_i - E_i)^2}{E_i}$$

- where C_A is the count (not the probability) of letter A, and E_A is the expected count of A.

Cryptanalysis Using Chi-squared Statistics

- **Example:** Ciphertext:

aoljhlzhjpwolypzvlvmaollhysplzaruvduhukzptwslzajpwolyzpzahfwlv
zbizapabapvujpwolypudopjohljoslaalypuaolwshpualeapzzopmalkhijyahpu
ubtilyvmwshjlkvdualhswohila

- Step1: Calculate the frequency of each 25 possible keys (A, B, C,,Z).
- Step2: Calculate the Chi-squared statistic for each key and find the lowest that is the key.

- The cipher shown above is 162 characters
- Letter A appears 18 times in the cipher ($C_A=18$)
- Expected values of A is $162 \times 0.082 = 13.284 = E_A$

- The chi-squared statistic is
$$\frac{(18 - 13.284)^2}{13.284} = 1.674$$

A	18
B	3
C	0
D	3
E	1
F	1
G	0
H	12
I	3
J	8
K	3
L	22
M	4
N	0
O	12
P	17
Q	0
R	1
S	6
T	2
U	11
V	7
W	8
X	0
Y	8
Z	12

Cryptanalysis Using Chi-squared Statistics

We also need to perform this procedure for the other letters, then add the 26 results that we have. The result of this is around 1634.09. To find the correct key we have to do this for each key.

decryption key	plaintext	chi-squared
0	AOLJHLZHYJPWOLYPZVULVMAOL ...	1634.09
1	ZNKIGYGYXIOVKNKXOYUTKULZNK ...	3441.13
2	YMHFXFWHNUMJWNXTSJTKYMJ ...	2973.71
3	XLIGEIVEGMTLIVMWSRISJXLI ...	1551.67
4	WKHFDHVDUFLSKHULVRQRIWKH ...	1199.40
5	VJGECGUCTEKRJGTUQPQGVJG ...	1466.62
6	UIFDBFTBDSJQIFSTJPOPPGUIF ...	1782.26
7	THECAESARCIPHERISONEOF THE ...	33.67
8	SGDBZDRZQBHOGDQHRNMDNESGD ...	1747.07
9	RFCAYCQYPAGNFCPGQMLCMDRFC ...	1386.62
10	QEBZXPXOZFMEBOFPLKBLCQEB ...	3423.96
11	PDAYNAOWNVELDANEOKJAKBPDA ...	809.38
12	OCZXVZNVXMKCZMDNJIZJAO CZ ...	4646.96
13	NBYWUYMULWCJBYLCMIHYIZNBY ...	724.11
14	MAXVTXLT KVBIAXKBLHGXYMAX ...	2159.43
15	LZWUSWKSJUAHZWJAKGFWGLZW ...	1787.26
16	KYVTRVJRITZGYVIZJFEVFWKYV ...	3527.17
17	JXUSQIUIQHSYFXUHYTIEDUEVJXU ...	2967.66
18	IWTRPTHPRXWXTGXHDCDUIWT ...	1368.70
19	HVSQSGOFQMDVSVFWGCBSCTHVS ...	929.17
20	GURPNRENPVCUREVFBARBSGUR ...	461.19
21	FTQOMEMDOUBTQDUEAZQARFTQ ...	4395.68
22	ESPNLPDLCNTASPTCDZYPZQESP ...	703.43
23	DROMKOCKBMSZROBSXYOYPDRO ...	1226.79
24	CQNLJNBJALRYQNARBXWXXOCQN ...	1817.85
25	BPMKIMAIZKQXPMZQAWMWNBP M ...	2939.16

the key used to encipher the message was 7

Find the Key of Monoalphabetic Substitution Cipher

GFS WMY OG LGDV\$ MF SENKYHOSU ESLLMRS, PC WS BFGW POL DMFRQMRS, PL OG CPFU M UP
HDMPFOSXO GC OIS LMES DMFRQMRS DGFR SFGQRI OG CPDD GFS LISSO GK LG, MFU OISF WS NC
GNNQKKSFNSL GC SMNI DSOOSK. WS NMDD OIS EGLO CKSJQSFODY GNNQKKPFR DSOOSK OIS 'CPKLO',
EGLO GNNQKKPFR DSOOSK OIS 'LSNGFU' OIS CGDDGWPFR EGLO GNNQKKPFR DSOOSK OIS 'OIPKU', MF
QFOPD WS MNNGQFO CGK MDD OIS UPCCSKSFO DSOOSK PF OIS HDMPFOSXO LMEHDS. OISF WS DGG
NPHISK OSXO WS WMFO OG LGDV\$ MFU WS MDLG NDMLPCY POL LYEAGDL. WS CPFU OIS EGLO GNN
LYEAGD MFU NIMFR\$ PO OG OIS CGKE GC OIS 'CPKLO' DSOOSK GC OIS HDMPFOSXO LMEHDS, OIS FS
NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'LSNGFU' DSOOSK, MFU OIS CGDDGWPFR EGLO
LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'OIPKU' DSOOSK, MFU LG GF, QFOPD WS MNNGQFO CGK MDD
GC OIS NKYHGRKME WS WMFO OG LGDV\$.

The frequency of each letter appearing in the above intercept.

Ciphertext Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
Frequency	5	2	26	42	23	51	67	8	33	1	35	39	35	29	85	30	14	17	88	0	17	3	16	6	1

Find the Frequency of all the Letters Appearing in Intercept

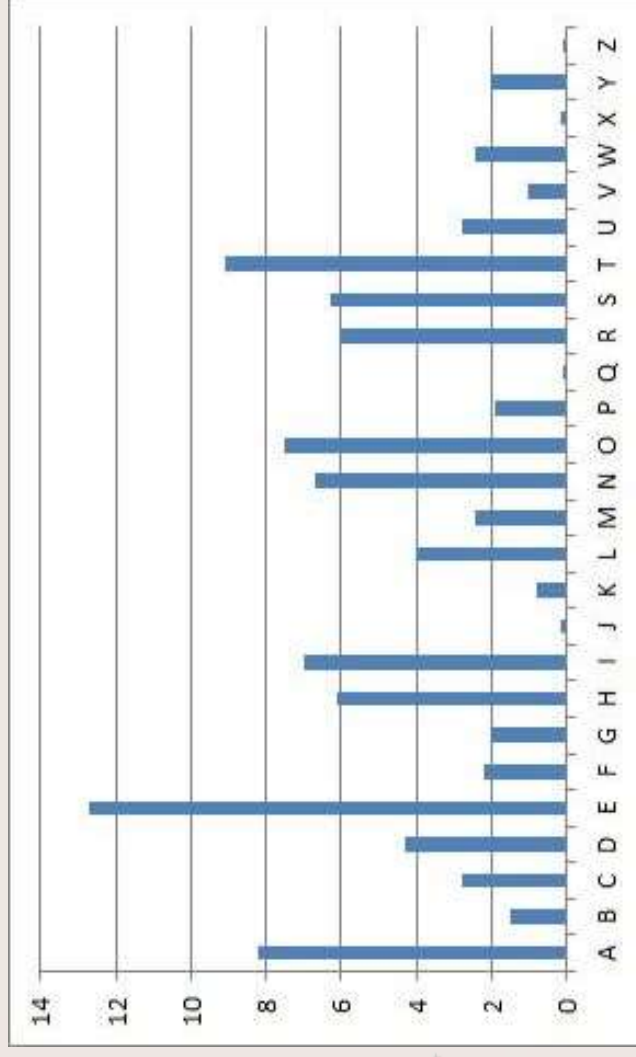
Ciphertext Letter	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Frequency	5	2	26	42	23	51	67	8	33	1	35	39	35	29	85	30	14	17	88	0	17	3	16	6

- The frequency of each letter appearing in the above intercept.

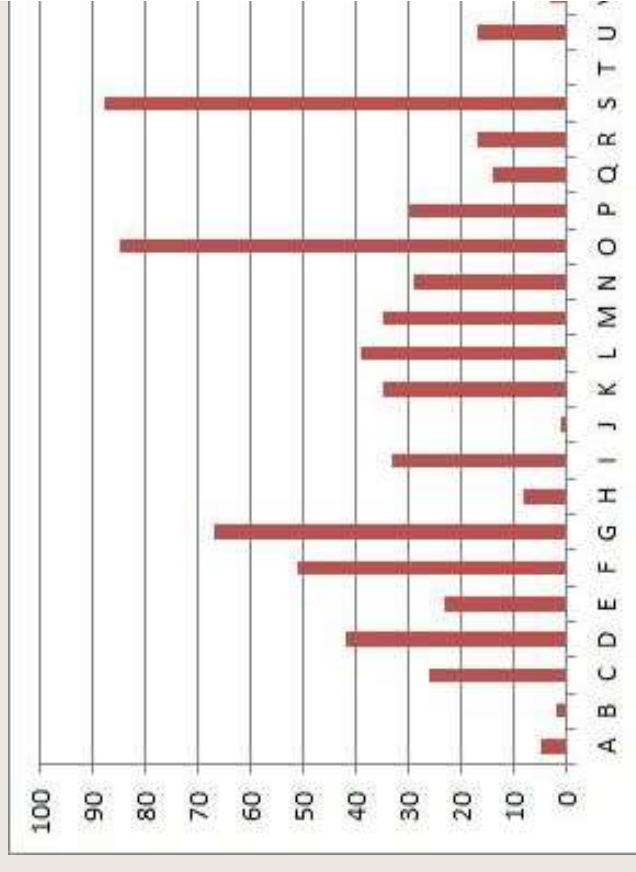
Ciphertext Letter	S	O	G	F	D	L	K	M	I	P	N	C	E	R	U	W	Q	Y	H	X	A	V	B	J
Frequency	88	85	67	51	42	39	35	35	33	30	29	26	23	17	17	16	14	10	8	6	5	3	2	1

- The above frequencies ordered from most common to least to make comparisons easier

Find the Frequency of all the Letters Appearing in Intercept



The Standard English Letter Frequencies.



The letter frequencies of the letters in the ciphertext.

- We see that the most common ciphertext letter is "S", closely followed by "O". From the chart above, we can guess that these two letters represent "e" and "t" respectively.

Decrypting without key

GFS WMY OG LGDV\$ MF SFNKYHOSU ESLLMRS, PC WS BFGW POL DMFRQMRS, PL OG CPFU M UP
HDMPFOSXO GC OIS LMES DMFRQMRS DGFR SFGQRI OG CPDD GFS LISSO GK LG, MFU OISF WS NC
GNNQKKSFNSL GC SMNI DSOOSK. WS NMDD OIS EGLO CKSJQSFODY GNNQKKPFR DSOOSK OIS 'CPKLO',
EGLO GNNQKKPFR DSOOSK OIS 'LSNGFU' OIS CGDDGWPFR EGLO GNNQKKPFR DSOOSK OIS 'OIPKU', MF
QFOPD WS MNNGQFO CGK MDD OIS UPCCSKSFO DSOOSK PF OIS HDMPFOSXO LMEHDS. OISF WS DGG
NPHISK OSXO WS WMFO OG LGDV\$ MFU WS MDLG NDMLPCY POL LYEAGDL. WS CPFU OIS EGLO GNN
LYEAGD MFU NIMFR\$ PO OG OIS CGKE GC OIS 'CPKLO' DSOOSK GC OIS HDMPFOSXO LMEHDS, OIS FS
NGEEGF LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'LSNGFU' DSOOSK, MFU OIS CGDDGWPFR EGLO
LYEAGD PL NIMFRSU OG OIS CGKE GC OIS 'OIPKU' DSOOSK, MFU LG GF, QFOPD WS MNNGQFO CGK MDD
GC OIS NKYHGRKME WS WMFO OG LGDV\$.

After making these substitutions S □ e and O □ t we get:

GFe WMY tG LGDVe MF eFNKYHteU EeLLMRe, PC We BFGW PtL DMFRQMRe, PL tG CPFU M UPCCeKeFt HDMPFt
LMEE DMFRQMRe DGFR eFGQRI tG CPDD GFe Lleet GK LG, MFU tLeF We NGQFt tle GNNQKKeFNeL GC eMNI
NMDD tle EGLt CKeJQeFtDY GNNQKKPFR DetteK tle 'CPKLt', tle FeXt EGLt GNNQKKPFR DetteK tle 'Le
CGDDGWPFR EGLt GNNQKKPFR DetteK tle 'tIPKU', MFU LG GF, QFtPD We MNNGQFt CGK MDD tle UPCCeKeFt D
tle HDMPFteXt LMEHDe. tLeF We DGGb Mt tle NPHLeK teXt We WMFt tG LGDVe MFU We MDLG NDMLPCY PtL L
We CPFU tle EGLt GNNQKKPFR LYEAGD MFU NIMFR\$ Pt tG tle CGKE GC tle 'CPKLt' DetteK GC tle HDMPFteXt
tle FeXt EGLt NGEeGF LYEAGD PL NIMFR\$U tG tle CGKE GC tle 'LeNGFU' DetteK, MFU tle CGDDGWPFR EGLt
LYEAGD PL NIMFR\$U tG tle CGKE GC tle 'tIPKU' DetteK, MFU LG GF, QFtPD We MNNGQFt CGK MDD LYEAGH
NKYHtGRKME We WMFt tG LGDVe.

Decrypting without key

*GFe WMY tG LGDVe MF eFNKYHteU EeLLMRe, PC We BFGW PtL DMFRQMRe, PL tG CPFU M UPQ
HDMPFteXt GC tle LMEe DMFRQMRe DGFR eFGQRI tG CPDD GFe Lleet GK LG, MFU tleF We N
GNNQKKeFNeL GC eMNI DetteK. We NMDD tle EGLt CKeJQeFtDY GNNQKKPFR DetteK tle 'CPKlt', MFU
EGLt GNNQKKPFR DetteK tle 'LeNGFU' tle CGDDGWPFER EGLt GNNQKKPFR DetteK tle 'tIPKU', MFU
QFtPD We MNNGQFt CGK MDD tle UPCCeKeFt DetteKL PF tle HDMPFteXt LMEHDe. tleF We DGG
NPHIeK teXt We WMFt tG LGDVe MFU We MDLG NDMLLPcY PtL LYEAGDL. We CPFU tle EGLt GNNQ
LYEAGD MFU NIMFRe Pt tG tle CGKE GC tle 'CPKlt' DetteK GC tle HDMPFteXt LMEHDe, tle F
NGEEGF LYEAGD PL NIMFReU tG tle CGKE GC tle 'LeNGFU' DetteK, MFU tle CGDDGWPFER EGLt
LYEAGD PL NIMFReU tG tle CGKE GC tle 'tIPKU' DetteK, MFU LG GF, QFtPD We MNNGQFt C
LYEAGDL GC tle NKYHtGRKME We WMFt tG LGDVe.*

- We now notice that the word "tle" is appearing frequently in the passage. In English, the common 3 letter word is "the" and this fits with what we have already done, which suggests that "I" should be decrypted to "h".
- Also, by looking at the frequencies again, we see the next most common letter is "G", probably one of "a", "i" or "o". We see that the third word is "tG", and the only one of the options that makes sense is "to", so we guess "G" is "o".

Decrypting without key

oFe WMY to LoDVe MF eFNKYHteU EeLLMRe, PC We BFoW PtL DMFRQMRe, PL to CPFU M UPQ
HDMPFteXt oC the LMEe DMFRQMRe DoFR eFoQRh to CPDD *oFe* *Lheet* oK Lo, MFU theF We N
oNNQKKeFNeL oC eMNh DetteK. We NMDD the EoLt CKeJQeFtDY oNNQKKPFR DetteK the 'CPKLi',
EoLt oNNQKKPFR DetteK the 'LeNoFU' the CoDDoWPFR EoLt oNNQKKPFR DetteK the 'thPKU', MF
QFtPD We MNNNoQFt CoK MDD the UPCCeKeFt DetteKL PF the HDMPFteXt LMEHDe. theF We Doo
NPHheK teXt We WMFt to LoDVe MFU We MDLo NDMLLPCY PtL LYEAOdL. We CPFU the EoLt oNNQ
LYEAoD MFU NhMFRe Pt to the CoKE oC the 'CPKLi' DetteK oC the HDMPFteXt LMEHDe, the F
NoEEoF LYEAOd PL NhMFReU to the CoKE oC the 'LeNoFU' DetteK, MFU the CoDDoWPFR EoLt
LYEAoD PL NhMFReU to the CoKE oC the 'thPKU' DetteK, MFU Lo oF, QFtPD We MNNNoQFt Co
LYEAoDL oC the NKYHtoRKME We WMFt to LoDVe.

The first word is now "oFe", which when considered with the appearance of "theF", leads us to the conclu
"F" is "n". This also fits in with the frequencies of both letters in the tables.
Also see *Lheet* , which is most likely "sheet" so replace "L" with "s". Again, the frequencis of these two l
about right.

Decrypting without key

one WMY to soDVe Mn enNKYHteU EessMRe, PC We BnoW Pts DMnRQMRe, Ps to CPnU M UP
HDMPnteXt oC the sMEe DMnRQMRe DonR enoQRh to CPDD one sheet oK so, MnU then We N
oNNQKKenNes oC eMNh DetteK. We NMDD the Eost CKeJQentDY oNNQKKPnR DetteK the 'CPKst', the
oNNQKKPnR DetteK the 'seNonU' the CoDDoWPnR Eost oNNQKKPnR DetteK the 'thPKU', MnU so on, Q
MNNNoQnt CoK MDD the UPCCeKent DetteKs Pn the HDMPnteXt sMEHDe. then We DooB Mt the NPHhel
WMnt to soDVe MnU We MDso NDMssPCY Pts sYEAoDs. We CPnU the Eost oNNQKKPnR sYEAoD MnU
Pt to the CoKE oC the 'CPKst' DetteK oC the HDMPnteXt sMEHDe, the neXt Eost NoEEon sYEAoD Ps NhM
the CoKE oC the 'seNonU' DetteK, MnU the CoDDoWPnR Eost NoEEon sYEAoD Ps NhMnReU to the CoK
'thPKU' DetteK, MnU so on, QntPD We MNNNoQnt CoK MDD sYEAoDs oC the NKYHtoRKME We WMnt to

- We see the word "soDVe", which could be "solve", implying the transformations of "D" and "V" to "l" and "v" respectively.
- In the second line we now have the phrase "one sheet oK so", which suggests that "K" is "r".
-

Decrypting without key

one WMY to solve Mn enNrYHteU EessMRe, PC We BnoW Pts lMnRQMRe, Ps to CPnU M UP
HlMPnteXt oC the sMEe lMnRQMRe lonR enoQRh to CPll one sheet or so, MnU then We N
oNNQrrrenNes oC eMNH letter. We NMll the Eost CreJQently oNNQrrPnR letter the 'CPrst', t
Eost oNNQrrPnR letter the 'seNonU' the ColloWPnR Eost oNNQrrPnR letter the 'thPrU', Mn
QntPl We MNNoQnt Cor Mll the UPCCerent letters Pn the HlMPnteXt sMEHle. then We loo
NPHher teXt We WMnt to solve MnU We Mlso NIMssPCY Pts sYEAols. We CPnU the Eost oNNQ
sYEAol MnU NhMnRe Pt to the CorE oC the 'CPrst' letter oC the HlMPnteXt sMEHle, the n
NoEEon sYEAol Ps NhMnReU to the CorE oC the 'seNonU' letter, MnU the ColloWPnR Eost
sYEAol Ps NhMnReU to the CorE oC the 'thPrU' letter, MnU so on, QntPl We MNNoQnt
sYEAols oC the NrYHtoRrME We WMnt to solve.

In the middle of the second line we have the word "enoQRh", which is likely to be "enough", and so we have the transformations "Q" to "u" and "R" to "h" respectively.

one WMY to solve Mn enNrYHteU EessMge, PC We BnoW Pts lMnguMge, Ps to CPnU M UPCCerent HlMPnteXt oC the sMEe lMnguMge lo
CPll one sheet or so, MnU then We Nount the oNNurrenNes oC eMNh letter. We NMll the Eost CreJuently oNNurrPng letter the 'CPrst',
oNNurrPng letter the 'seNonU' the ColloWPng Eost oNNurrPng letter the 'thPrU', MnU so on, untPl We MNNount Cor Mll the UPCCerent
HlMPnteXt sMEHle. then We looB Mt the NPHher teXt We WMnt to solve MnU We Miso NIMssPCY Pts sYEAols. We CPnU the Eost oNNur
MnU NhMnge Pt to the CorE oC the 'CPrst' letter oC the HlMPnteXt sMEHle, the neXt Eost NoEEon sYEAol Ps NhMngeU to the CorE oC
letter, MnU the ColloWPng Eost NoEEon sYEAol Ps NhMngeU to the CorE oC the 'thPrU' letter, MnU so on, untPl We MNNount Cor Mll sY
NrYHtogrME We WMnt to solve.

We have the word "Nount" which is could be "count" and "EessMge" which is likely to be "message", giving u
"E" and "M" and "c", "m" and "a".

one WaY to solve an encrYHteU message, PC We BnoW Pts language, Ps to CPnU a UPCCerent HlaPnteXt oC the same lan
enough to CPll one sheet or so, anU then We count the occurrences oC each letter. We call the most CreJuently occurPr
'CPrst', the neXt most occurPng letter the 'seconU' the ColloWPng most occurPng letter the 'thPrU', anU so on, untPl We
all the UPCCerent letters Pn the HlaPnteXt samHle. then We looB at the cPHher teXt We Want to solve anU We also cl
sYmAols. We CPnU the most occurPng sYmAol anU change Pt to the Corm oC the 'CPrst' letter oC the HlaPnteXt samH
most common sYmAol Ps changeU to the Corm oC the 'seconU' letter, anU the ColloWPng most common sYmAol Ps cha
Corm oC the 'thPrU' letter, anU so on, untPl We account Cor all sYmAols oC the crYHtogram We Want to solve.

- It is likely that "W" -> "w", "X" -> "x", "Y" -> "y" and "Z" -> "z".
- The word "occurrPng" is clearly meant to read "occurring", and it is likely that "sYmAol" is "sy

one way to solve an encrypted message, is to know its language, is to find a cipher character in the language long enough to fill one sheet or so, and then we count the occurrences of each letter. we call the first occurring letter the 'first', the next most occurring letter the 'second' the following most occurring the 'third', and so on, until we account for all the cipher characters in the plaintext sample, then we look at the cipher text we want to solve and we also classify its symbols. we find the most common symbol and change it to the form of the 'first' letter of the plaintext sample, the next common symbol is changed to the form of the 'second' letter, and the following most common symbol is changed to the form of the 'third' letter, and so on, until we account for all symbols of the ciphertext we want to solve.

- We can now see that "C" is "f", "B" is "k", "U" is "d", "J" is "q" and "H" is "p".

one way to solve an encrypted message, if we know its language, is to find a different plaintext of the language long enough to fill one sheet or so, and then we count the occurrences of each letter. we find the most frequently occurring letter the 'first', the next most occurring letter the 'second' the following most occurring letter the 'third', and so on, until we account for all the different letters in the plaintext sample, then we look at the cipher text we want to solve and we also classify its symbols. we find the most common symbol and change it to the form of the 'first' letter of the plaintext sample, the next common symbol is changed to the form of the 'second' letter, and the following most common symbol is changed to the form of the 'third' letter, and so on, until we account for all symbols of the ciphertext we want to solve.

Reveals the keyword

- The reconstructed ciphertext alphabet reveals the keyword

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x
Ciphertext Alphabet	M	A	N	U	S	C	R	I	P	T	B	D	E	F	G	H	J	K	L	O	Q	V	W	X

Vigenère Cipher

- Let letters A-Z are taken to be number 0-25. Let $\{P_1, P_2, P_3, \dots, P_n \mid P_i \in \{A, \dots, Z\}\}$ and $K = \{K_1, K_2, K_3, \dots, K_m \mid K_i \in \{A, \dots, Z\}\}$ are the plaintext and key respectively. The Vigenère Cipher can be defined algebraically

- **Encryption** $C_i = E_k(P_i) = (P_i + K_{(i \bmod m)}) \bmod 26$
- **Decryption** $P_i = D_k(C_i) = (C_i - K_{(i \bmod m)}) \bmod 26$

Example

Plaintext: ATTACKATDAWN = {0, 19, 19, 0, 2, 10, 0, 19, 3, 0, 22, 13}

Key: LEMON = {11, 4, 12, 13, 14}

Plaintext (P)

Key(K)

$$(P_i + K_{(i \bmod m)}) \bmod 26$$

Cyphertext (C)

A	T	T	A	C	K	A	T	D	A	W
0	19	19	0	2	10	0	19	3	0	22
11	4	12	13	14	11	4	12	13	14	11
11	23	5	13	16	21	4	5	16	14	7
L	X	F	O	P	V	E	F	R	N	H

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Plaintext Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Vigenere Cipher

- **Plaintext:**

ATTACKATDAWN

- **Key:**

LEMON

- **Keystream:**

LEMONLEMONLE

- **Ciphertext:**

LXFOPVEFRNHR

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
A	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
B	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
C	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A
D	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B
E	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C
F	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D
G	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E
H	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F
I	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G
J	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H
K	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I
L	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J
M	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K
N	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L
O	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M
P	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Q	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
R	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
S	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
T	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
U	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
V	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
W	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
X	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
Y	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
Z	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X

Cryptanalysis of the Vigenere Cipher

- if very long keys are used the vigenere cipher can be unbreakable.
- But if short keys are used, or if we have a lot of cipher compared to the key length, the vigenere cipher is quite solvable.
- See this:

<https://crypto.interactive-maths.com/kasiski-analysis-breaking-the-code.html>

- <http://practicalcryptography.com/cryptanalysis/stochastic-searching/cryptanalysis-vigenere-cipher/>

Affine Cipher

- The 'key' for the Affine cipher consists of 2 numbers (a, b)
- Assumes the use of a 26 character alphabet ($m=26$).
- $GCD(a, m)=1$
- Affine Cipher can be defined algebraically
- Encryption $C = aP + b \pmod{m}$
- Decryption $P = (C - b)a^{-1} \pmod{m}$
- where a^{-1} is the multiplicative inverse of a in the group of integers modulo m
- Multiplicative inverse of a is x and
- $ax = 1 \pmod{m}$

Example

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	z
Plaintext Value	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

key (a=5, b=8)

Encrypt:

Plaintext	a	f	f	i	n	e	c	i	p	h	e	r	
x	0	5	5	8	13	4		2	8	15	7	4	17
5x+8	8	33	33	48	73	28		18	48	83	43	28	93
(5x+8) mod 26	8	7	7	22	21	2		18	22	5	17	2	15
Ciphertext	I	H	H	W	V	C		S	W	F	R	C	P

Decrypt: the inverse of 5 modulo 26 is 21, i.e. $5 \cdot 21 = 1 \pmod{26}$.

Ciphertext	I	H	H	W	V	C		S	W	F	R	C	P
y	8	7	7	22	21	2		18	22	5	17	2	15
$21(y - 8)$	0	-21	-21	294	273	-126		210	294	-63	189	-126	147
$21(y - 8) \bmod 26$	0	5	5	8	13	4		2	8	15	7	4	17
Plaintext	a	f	f	i	n	e		c	i	p	h	e	r

Cryptanalysis of Affine Cipher

The Affine cipher is a very insecure cipher, is a monoalphabetic substitution cipher, so methods that are used to cryptanalyze substitution ciphers can be used for the affine cipher. ciphers can also be cracked if any 2 characters are known.

Imagine we have a ciphertext in which h and q are most common letters. Therefore h and q are most common letters. Therefore h and q are most common letters. Therefore h and q are most common letters.

$$\begin{aligned}a \times 4 + b &= 7 \pmod{26} \\a \times 19 + b &= 16 \pmod{26}\end{aligned}$$

$$\begin{aligned}a \times p + b &= r \pmod{26} \\a \times q + b &= s \pmod{26}\end{aligned}$$

$$\begin{aligned}a &= D^{-1}(r - s) \pmod{26} \\b &= D^{-1}(ps - qr) \pmod{26}\end{aligned}$$

Where $D = p - q$ and D^{-1} is a number, x , such that $D \cdot x = 1 \pmod{26}$.

$p=4$, $r=7$, $q=19$, $s=16$. $D = p - q = 15 \pmod{26}$. This means $D^{-1} = 19$. So:

$$\begin{aligned}a &= 19 \times (r - s) \pmod{26} = 19 \times (-9) \pmod{26} = 11 \\b &= 19 \times (ps - qr) \pmod{26} = 19 \times (64 - 133) \pmod{26} = 15\end{aligned}$$

The key is (11, 15). Decrypt the cipher text if it is sensible fine otherwise try with different common letters.

For implementation: <http://practicalcryptography.com/ciphers/classical-era/affine/>

Cryptanalysis of Affine Cipher (second method)

The affine cipher has 2 key numbers, 'a' and 'b'. 'b' can range from 0 to 25, and 'a' can have any of 1,3,5,7,9,11,15,17,19,21,23,25. The number of possible keys is $12 \times 26 - 1 = 311$. We iterate over each of these combinations, determine the fitness of each combination, then choose the best.

Quadgram Statistics as a Fitness Measure:

For the text **ATTACK**, the quadgrams are **ATTA**, **TTAC**, and **TACK**. The total probability is

$$p(\text{ATTACK}) = p(\text{ATTA}) \times p(\text{TTAC}) \times p(\text{TACK})$$

where

$$p(\text{ATTA}) = \frac{\text{count}(\text{ATTA})}{N}$$

count() is the number of times the particular quadgram occurred in English text, N is the total number of quadgrams in the training sample.

When multiplying many small probabilities, numerical underflow can occur in floating point numbers. For this reason, the logarithm is taken of each probability and it is used as the fitness of the text.

$$\log(p(\text{ATTACK})) = \log(p(\text{ATTA})) + \log(p(\text{TTAC})) + \log(p(\text{TACK}))$$

A higher number means it is more likely to be English, while a lower number means it is less likely to be English.

For implementation: <http://practicalcryptography.com/cryptanalysis/text-characterisation/quadgrams/>

Cryptanalysis of Affine Cipher (second method)

Ciphertext

QUVNLAUVILZKVZZZVNHIVQUFSFZHWZQLQHQLJSNLAUVIFZLQLZYZHKS VIFWKVQJ
QUVFQQFNTZQUFQPJITFDFLSZQZHWZQLQHQLJSNLAUVIZLSFEELQLJSQJJQUVIFQQH

decrypting with all possible keys (only a few are shown in the table):

a	b	plaintext	fitness
(1, 1)		PTUMKZTUHKYJUYYUMGH...	-1144.91
(3, 5)		VFOUCHFOBCYTOYYOUSB...	-1112.00
(11, 18)		OMFJXWMSXDEFDDDFJZS...	-1141.52
(17, 5)		THECIPHERISLESSECUR...	-593.58
(21, 1)		XRWIVVRWJYQTWQQWIEJ...	-1194.72
(25, 9)		TPOWYJPOBYKZOKKOWCB...	-1117.56

5, 5) and plaintext is

THECIPHERISLESSECURETHANASUBSTITUTIONCIPHERAS
ISVULNERABLETOALLOFTHEATTACKSTHATWORKAGAINSTSU
TITUTIONCIPHERSINADDITIONTOOTHERATTACKS

Hill Cipher

Invented by Lester S. Hill in 1929, the Hill cipher is a polygraphic substitution cipher based on linear algebra. it can work on digraphs, trigraphs (3 letter blocks) or theoretically any sized blocks.

Encryption: $c = Kp \pmod{26}$

Decryption: $p = K^{-1}c \pmod{26}$

- plaintext : $p \in \{0, 1, 2, 3, \dots, 25\}$
- ciphertext : $c \in \{0, 1, 2, 3, \dots, 25\}$
- key : K is an invertible matrix

Example

- Encrypt: “shortexample” with key HILL
- Key k: $\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$
- Plaintext: $\begin{pmatrix} s \\ h \end{pmatrix} \begin{pmatrix} o \\ r \end{pmatrix} \begin{pmatrix} t \\ e \end{pmatrix} \begin{pmatrix} x \\ a \end{pmatrix} \begin{pmatrix} m \\ p \end{pmatrix} \begin{pmatrix} l \\ e \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix}$
- Encryption:
$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} s \\ h \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} 182 \\ 275 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \end{pmatrix} \text{ mod } 26 = \begin{pmatrix} A \\ P \end{pmatrix}$$
- Ciphertext of "APADJ TFTWLFJ".
- Similarly do for rest vectors.

Example conti..

- To decrypt a ciphertext encoded using the Hill Cipher, we must find the inverse matrix.
- General method to calculate the inverse key matrix.
$$K^{-1} = d^{-1} \times adj(K)$$
- $K = \begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$
- The determinant of the key matrix.
$$\begin{vmatrix} 7 & 8 \\ 11 & 11 \end{vmatrix} = 7 \times 11 - 8 \times 11 = -11 = 15 \text{ mod } 26$$
- Multiplicative inverse of the determinant working modulo 26.
$$15 \times 7 = 105 = 1 \text{ mod } 26$$
- Find the adjoint matrix in module 26
$$adj \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$
$$adj \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} = \begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix} = \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix}$$
- K^{-1} :

$$7 \times \begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} = \begin{pmatrix} 77 & 126 \\ 105 & 49 \end{pmatrix} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \text{ mod } 26$$

$$\text{if } K = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}, \text{ then } K^{-1} = \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix}$$

Encryption

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix}$$

$$7 \times 18 + 8 \times 7 = 182$$

$$11 \times 18 + 11 \times 7 = 275$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} 182 \\ 275 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} 182 \\ 275 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} s \\ h \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} 182 \\ 275 \end{pmatrix} = \begin{pmatrix} 0 \\ 15 \end{pmatrix} \pmod{26} = \begin{pmatrix} A \\ P \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix}$$

$$7 \times 23 + 8 \times 0 = 161$$

$$11 \times 23 + 11 \times 0 = 253$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix} = \begin{pmatrix} 161 \\ 253 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix} = \begin{pmatrix} 161 \\ 253 \end{pmatrix} = \begin{pmatrix} 5 \\ 19 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} x \\ a \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 23 \\ 0 \end{pmatrix} = \begin{pmatrix} 161 \\ 253 \end{pmatrix} = \begin{pmatrix} 5 \\ 19 \end{pmatrix} \pmod{26} = \begin{pmatrix} F \\ T \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix}$$

$$7 \times 14 + 8 \times 17 = 234$$

$$11 \times 14 + 11 \times 17 = 341$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix} = \begin{pmatrix} 234 \\ 341 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix} = \begin{pmatrix} 234 \\ 341 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} o \\ r \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 14 \\ 17 \end{pmatrix} = \begin{pmatrix} 234 \\ 341 \end{pmatrix} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} \pmod{26} = \begin{pmatrix} A \\ D \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix}$$

$$7 \times 12 + 8 \times 15 = 204$$

$$11 \times 12 + 11 \times 15 = 297$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix} = \begin{pmatrix} 204 \\ 297 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix} = \begin{pmatrix} 204 \\ 297 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} m \\ p \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 12 \\ 15 \end{pmatrix} = \begin{pmatrix} 204 \\ 297 \end{pmatrix} = \begin{pmatrix} 22 \\ 11 \end{pmatrix} \pmod{26} = \begin{pmatrix} W \\ L \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix}$$

$$7 \times 19 + 8 \times 4 = 165$$

$$11 \times 19 + 11 \times 4 = 253$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix} = \begin{pmatrix} 165 \\ 253 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix} = \begin{pmatrix} 165 \\ 253 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} t \\ e \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 19 \\ 4 \end{pmatrix} = \begin{pmatrix} 165 \\ 253 \end{pmatrix} = \begin{pmatrix} 9 \\ 19 \end{pmatrix} \pmod{26} = \begin{pmatrix} J \\ T \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix}$$

$$7 \times 11 + 8 \times 4 = 109$$

$$11 \times 11 + 11 \times 4 = 165$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix} = \begin{pmatrix} 109 \\ 165 \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix} = \begin{pmatrix} 109 \\ 165 \end{pmatrix} \pmod{26}$$

$$\begin{pmatrix} H & I \\ L & L \end{pmatrix} \begin{pmatrix} l \\ e \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \end{pmatrix} = \begin{pmatrix} 109 \\ 165 \end{pmatrix} = \begin{pmatrix} 5 \\ 9 \end{pmatrix} \pmod{26} = \begin{pmatrix} F \\ T \end{pmatrix}$$

Decryption

ciphertext is "APADJ TFTWLFJ".

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} A \\ P \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 0 \\ 15 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 0 + 22 \times 15 \\ 1 \times 0 + 23 \times 15 \end{pmatrix} \\ &= \begin{pmatrix} 330 \\ 345 \end{pmatrix} \\ &= \begin{pmatrix} 18 \\ 7 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} s \\ h \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 0 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 0 + 22 \times 3 \\ 1 \times 0 + 23 \times 3 \end{pmatrix} \\ &= \begin{pmatrix} 66 \\ 69 \end{pmatrix} \\ &= \begin{pmatrix} 14 \\ 17 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} o \\ r \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} J \\ T \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 9 \\ 19 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 9 + 22 \times 19 \\ 1 \times 9 + 23 \times 19 \end{pmatrix} \\ &= \begin{pmatrix} 643 \\ 446 \end{pmatrix} \\ &= \begin{pmatrix} 19 \\ 4 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} t \\ e \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} F \\ T \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 5 \\ 19 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 5 + 22 \times 19 \\ 1 \times 5 + 23 \times 19 \end{pmatrix} \\ &= \begin{pmatrix} 543 \\ 442 \end{pmatrix} \\ &= \begin{pmatrix} 23 \\ 0 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} x \\ a \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} W \\ L \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 22 \\ 11 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 22 + 22 \times 11 \\ 1 \times 22 + 23 \times 11 \end{pmatrix} \\ &= \begin{pmatrix} 792 \\ 275 \end{pmatrix} \\ &= \begin{pmatrix} 12 \\ 15 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} m \\ p \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} F \\ J \end{pmatrix} &= \begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 5 \\ 9 \end{pmatrix} \\ &= \begin{pmatrix} 25 \times 5 + 22 \times 9 \\ 1 \times 5 + 23 \times 9 \end{pmatrix} \\ &= \begin{pmatrix} 323 \\ 212 \end{pmatrix} \\ &= \begin{pmatrix} 11 \\ 4 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} l \\ e \end{pmatrix}\end{aligned}$$

3x3 Example - Encryption

- Key phrase “**BACKUP**” into a matrix. key phrase is a few letters short, Use **A, B, C**.
- Encrypt “**retreat now**” using the key phrase *back up* and a 3 x 3 matrix. Use some nulls make the plaintext the right length.
- We perform all the matrix multiplications, and take the column vectors modulo 26.

$$\begin{pmatrix} B & A & C \\ K & U & P \\ A & B & C \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 2 \\ 10 & 20 & 15 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} r \\ e \\ t \end{pmatrix} \begin{pmatrix} r \\ e \\ a \end{pmatrix} \begin{pmatrix} t \\ n \\ o \end{pmatrix} \begin{pmatrix} w \\ x \\ x \end{pmatrix}$$

$$\begin{pmatrix} 17 \\ 4 \\ 19 \end{pmatrix} \begin{pmatrix} 17 \\ 4 \\ 0 \end{pmatrix} \begin{pmatrix} 19 \\ 13 \\ 14 \end{pmatrix} \begin{pmatrix} 22 \\ 23 \\ 23 \end{pmatrix}$$

$$\begin{pmatrix} B & A & C \\ K & U & P \\ A & B & C \end{pmatrix} \begin{pmatrix} r \\ e \\ t \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 10 & 20 & 15 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 17 \\ 4 \\ 19 \end{pmatrix} \\ = \begin{pmatrix} 1 \times 17 + 0 \times 4 + 2 \times 19 \\ 10 \times 17 + 20 \times 4 + 15 \times 19 \\ 0 \times 17 + 1 \times 4 + 2 \times 19 \end{pmatrix} \\ = \begin{pmatrix} 55 \\ 535 \\ 42 \end{pmatrix} \mod 26 \\ = \begin{pmatrix} 3 \\ 15 \\ 16 \end{pmatrix} \begin{pmatrix} D \\ P \\ Q \end{pmatrix}$$

$$\begin{pmatrix} B & A & C \\ K & U & P \\ A & B & C \end{pmatrix} \begin{pmatrix} r \\ e \\ a \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 10 & 20 & 15 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 17 \\ 4 \\ 0 \end{pmatrix} \\ = \begin{pmatrix} 1 \times 17 + 0 \times 4 + 2 \times 0 \\ 10 \times 17 + 20 \times 4 + 15 \times 0 \\ 0 \times 17 + 1 \times 4 + 2 \times 0 \end{pmatrix} \\ = \begin{pmatrix} 17 \\ 250 \\ 4 \end{pmatrix} \mod 26 \\ = \begin{pmatrix} 17 \\ 16 \\ 4 \end{pmatrix} \begin{pmatrix} R \\ Q \\ E \end{pmatrix}$$

$$\begin{pmatrix} B & A & C \\ K & U & P \\ A & B & C \end{pmatrix} \begin{pmatrix} t \\ n \\ o \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 10 & 20 & 15 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 19 \\ 13 \\ 14 \end{pmatrix} \\ = \begin{pmatrix} 1 \times 19 + 0 \times 13 + 2 \times 14 \\ 10 \times 19 + 20 \times 13 + 15 \times 14 \\ 0 \times 19 + 1 \times 13 + 2 \times 14 \end{pmatrix} \\ = \begin{pmatrix} 47 \\ 660 \\ 41 \end{pmatrix} \mod 26 \\ = \begin{pmatrix} 21 \\ 10 \\ 15 \end{pmatrix} \begin{pmatrix} V \\ K \\ P \end{pmatrix}$$

$$\begin{pmatrix} B & A & C \\ K & U & P \\ A & B & C \end{pmatrix} \begin{pmatrix} w \\ x \\ x \end{pmatrix} = \begin{pmatrix} 1 & 0 & 2 \\ 10 & 20 & 15 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 22 \\ 23 \\ 23 \end{pmatrix} \\ = \begin{pmatrix} 1 \times 22 + 0 \times 23 + \\ 10 \times 22 + 20 \times 23 + \\ 0 \times 22 + 1 \times 23 + \end{pmatrix} \\ = \begin{pmatrix} 68 \\ 1025 \\ 69 \end{pmatrix} \mod 26 \\ = \begin{pmatrix} 16 \\ 11 \\ 17 \end{pmatrix} \begin{pmatrix} Q \\ L \\ R \end{pmatrix}$$

- ciphertext of "DPQRQ EVKPQ LR".

3x3 Example - Decryption

- Ciphertext message "SYICHOLER"

$$\begin{pmatrix} A & L & P \\ H & A & B \\ E & T & A \end{pmatrix}$$

$$\begin{pmatrix} 0 & 11 & 15 \\ 7 & 0 & 1 \\ 4 & 19 & 0 \end{pmatrix}$$

- Let key= *ALPHABET*.

- Step 1 - Find the Multiplicative Inverse of the Determinant of the above matrix
Finding the determinant of the 3 x 3 matrix with keyword alphabet using mod 26.

$$\begin{vmatrix} 0 & 11 & 15 \\ 7 & 0 & 1 \\ 4 & 19 & 0 \end{vmatrix} = 0 \begin{vmatrix} 0 & 1 \\ 19 & 0 \end{vmatrix} - 11 \begin{vmatrix} 7 & 1 \\ 4 & 19 \end{vmatrix} + 15 \begin{vmatrix} 7 & 0 \\ 4 & 19 \end{vmatrix} \\ = 0(0 - 19) - 11(0 - 4) + 15(133 - 0) \\ = 0 + 44 + 1995 \\ = 2039 \\ = 11 \pmod{26}$$

The multiplicative inverse is the number we multiply 11 by to get 1 modulo 26.

$$11 \times x = 1 \pmod{26}$$

$$dd^{-1} = 1 \pmod{26}$$

$$11 \times 19 = 209 = 1 \pmod{26}$$

multiplicative inverse of the determinant modulo 26 is 19

3x3 Example - Decryption

- Step 2 - Find the Adjugate Matrix

$$\begin{aligned} \text{adj} \begin{pmatrix} 0 & 11 & 15 \\ 7 & 4 & 19 \\ 4 & 19 & 0 \end{pmatrix} &= \begin{pmatrix} + \begin{vmatrix} 19 & 0 \\ 7 & 19 \end{vmatrix} & - \begin{vmatrix} 11 & 15 \\ 4 & 19 \end{vmatrix} & + \begin{vmatrix} 11 & 15 \\ 0 & 1 \end{vmatrix} \\ - \begin{vmatrix} 7 & 1 \\ 4 & 0 \end{vmatrix} & + \begin{vmatrix} 0 & 15 \\ 4 & 0 \end{vmatrix} & - \begin{vmatrix} 0 & 15 \\ 7 & 1 \end{vmatrix} \\ + \begin{vmatrix} 7 & 0 \\ 4 & 19 \end{vmatrix} & - \begin{vmatrix} 0 & 11 \\ 4 & 19 \end{vmatrix} & + \begin{vmatrix} 0 & 11 \\ 7 & 0 \end{vmatrix} \end{pmatrix} \\ &= \begin{pmatrix} -19 & 285 & 11 \\ 4 & -60 & 105 \\ 133 & 44 & -77 \end{pmatrix} \\ &= \begin{pmatrix} 7 & 25 & 11 \\ 4 & 18 & 1 \\ 3 & 18 & 1 \end{pmatrix} \text{ mod } 26 \end{aligned}$$

- Step 3 - Multiply the Multiplicative Inverse Determinant by the Adjugate Matrix

$$19 \times \begin{pmatrix} 7 & 25 & 11 \\ 4 & 18 & 1 \\ 3 & 18 & 1 \end{pmatrix} = \begin{pmatrix} 133 & 475 & 209 \\ 76 & 342 & 19 \\ 57 & 342 & 19 \end{pmatrix} = \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \text{ mod } 26$$

$$\text{if } K = \begin{pmatrix} 0 & 11 & 15 \\ 7 & 0 & 1 \\ 4 & 19 & 0 \end{pmatrix}, \text{ then } K^{-1} = \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix} \begin{pmatrix} S \\ Y \\ I \end{pmatrix} &= \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 18 \\ 24 \\ 8 \end{pmatrix} \\ &= \begin{pmatrix} 3 \times 18 + 7 \times 24 + 1 \times 8 \\ 24 \times 18 + 4 \times 24 + 19 \times 8 \\ 5 \times 18 + 4 \times 24 + 19 \times 8 \end{pmatrix} \\ &= \begin{pmatrix} 230 \\ 680 \\ 338 \end{pmatrix} \\ &= \begin{pmatrix} 22 \\ 4 \\ 0 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} w \\ e \\ a \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 1 \\ 4 \\ 0 \end{pmatrix} \begin{pmatrix} C \\ H \\ O \end{pmatrix} &= \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 2 \\ 7 \\ 14 \end{pmatrix} \\ &= \begin{pmatrix} 3 \times 2 + 7 \times 7 + 1 \times 14 \\ 24 \times 2 + 4 \times 7 + 19 \times 14 \\ 5 \times 2 + 4 \times 7 + 19 \times 14 \end{pmatrix} \\ &= \begin{pmatrix} 69 \\ 342 \\ 304 \end{pmatrix} \\ &= \begin{pmatrix} 17 \\ 4 \\ 18 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} r \\ e \\ s \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 1 \\ 19 \\ 4 \end{pmatrix} \begin{pmatrix} L \\ E \\ R \end{pmatrix} &= \begin{pmatrix} 3 & 7 & 1 \\ 24 & 4 & 19 \\ 5 & 4 & 19 \end{pmatrix} \begin{pmatrix} 11 \\ 4 \\ 17 \end{pmatrix} \\ &= \begin{pmatrix} 3 \times 11 + 7 \times 4 + 1 \times 17 \\ 24 \times 11 + 4 \times 4 + 19 \times 17 \\ 5 \times 11 + 4 \times 4 + 19 \times 17 \end{pmatrix} \\ &= \begin{pmatrix} 78 \\ 603 \\ 394 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 5 \\ 4 \end{pmatrix} \text{ mod } 26 \\ &= \begin{pmatrix} a \\ f \\ c \end{pmatrix} \end{aligned}$$

plaintext of "we are safe".

Cryptanalysis of Hill Cipher

- Ciphertext only attack is difficult
- known plaintext attack
- $$\begin{bmatrix} k_0 & k_1 \\ k_2 & k_3 \end{bmatrix} \times \begin{bmatrix} 18 & 14 \\ 7 & 17 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 15 & 3 \end{bmatrix} \pmod{26}$$
- $18k_0 + 7k_1 = 0$
- $14k_0 + 17k_1 = 0$
- $18k_2 + 7k_3 = 0$
- $14k_2 + 17k_3 = 0$
- solve to get the key.

Playfair Cipher

- **The Key:** A word transformed in 5x5 matrix (key square) and use sequence of 25 letters repeats. no 'j', it is combined with 'i'.
- **Example:**
- Key: 'playfair example', key square is

P	L	A	Y	F ^A
I	R	E	X ^A	M ^{PLE A}
B	C	D ^{EF}	G	H ^{I=J}
K ^{LM}	N	O	P ^R	S
T	U	V	W ^{XY}	Z
- Encrypting the message "Hide the gold in the tree stump" ("E" is repeated in tree)
- **Encryption :**
 - Second occurrence of letter replaces with an 'X'
 - Make even length plaintext appending 'X' at last if needed.
 - Make bigrams: HI DE TH EG OL DI NT HE TR EX ES TU MP

Playfair Cipher

- Bigram is in different rows and columns (shape: rectangle)-Pick same rows, opposite corners
- Assume one wants to encrypt the bigram OR.
- OR $\rightarrow$ ZX

Z	*	*	O	*
*	*	*	*	*
*	*	*	*	*
R	*	*	X	*
*	*	*	*	*

- In same rows (shape: row)-Pick item to right of each letter, wrap to left if needed.
- OR $\rightarrow$ YZ

*	*	*	*	*
*	O	Y	R	Z
*	*	*	*	*
*	*	*	*	*
*	*	*	*	*

- In same column (shape: column)-Pick item to below each letter, wrap to top if needed.
- OR $\rightarrow$ BY

*	*	O	*	*
*	*	B	*	*
*	*	*	*	*
*	*	*	R	*
*	*	*	Y	*

-

-

Four-Square Cipher

- Four-square cipher uses four 5 by 5 (5x5) matrices.
- Keywords "**example**" and "**keywords**"
- **Key:** Four-square Matrices
-

a	b	c	d	e	E	X	A	M	P
f	g	h	i	k	L _E	B	C	D	F
l	m	n	o	p	G	H	I	K	N
q	p	r	s	t	O	Q	R	S	T
v	w	x	y	z	U	V	W	Y	Z
K	E	Y	W	O	a	b	c	d	e
R	D _S	A	B	C	f	g	h	i	j
F	G	H	I	L	k	l	m	n	o
M	N	P	Q	S	p	r	s	t	u
T	U	V	X	Z	v	w	x	y	z

- The plaintext matrices are in lowercase and the ciphertext matrices are in caps

Four-Square Cipher

- Encrypting the message "Hide the gold in the tree stump"
- Make digraphs: HI DE TH EG OL DI NT HE TR EE ES TU MP (add "X" at end if needed)
- Plaintext Digraph HI

a	b	c	d	e	E	X	A	M	P
f	g	h	i	k	L _E	B	C	D	F
l	m	n	o	p	G	H	I	K	N
q	p	r	s	t	O	Q	R	S	T
v	w	x	y	z	U	V	W	Y	Z
K	E	Y	W	O	a	b	c	d	e
R	D _S	A	B	C	f	q	h	i	j
F	G	H	I	L	k	l	m	n	o
M	N	P	Q	S	p	r	s	t	u
T	U	V	X	Z	v	w	x	y	z

raph DA

- Ciphertext: DAPWRBXC GIMBKP FYQQP SPQLIZ
- Decryption: Reverse of encryption

Simple Transposition Ciphers

- The letters or words of the plaintext are reordered in some way, fixed by a given rule (the key).
- "a simple example" becomes "ELPMAXE ELPMIS A"
- "a simple example" becomes "A ELPMIS ELPMAXE"

Double Transposition Cipher

Encrypt: Arrange the plain text into appropriate size matrix and then permute the row and according to specified permutations.

Decrypt: Undo the permutation as per key to recover the plaintext.

Key: size matrix and row and column permutation

Plaintext: 'attackatdawn' in to 3x4 matrix

$$\begin{bmatrix} a & t & t & a \\ c & k & a & t \\ d & a & w & n \end{bmatrix}$$

permute row (1, 2, 3) $\rightarrow$ (3, 2, 1) then permute the column (1, 2, 3, 4) $\rightarrow$ (4, 2, 1, 3)

$$\begin{bmatrix} a & t & t & a \\ c & k & a & t \\ d & a & w & n \end{bmatrix} \rightarrow \begin{bmatrix} d & a & w & n \\ c & k & a & t \\ a & t & t & a \end{bmatrix} \rightarrow \begin{bmatrix} n & a & d & w \\ t & k & c & a \\ a & t & a & t \end{bmatrix}$$

Ciphertext: NADWTKCAATAT

Double Transposition Cipher

● Ciphertext: NADWTKCAATAT

Key: (3x4) , row (1, 2, 3) $\rightarrow$ (3, 2, 1), column (1, 2, 3, 4) $\rightarrow$ (4, 2, 1, 3)

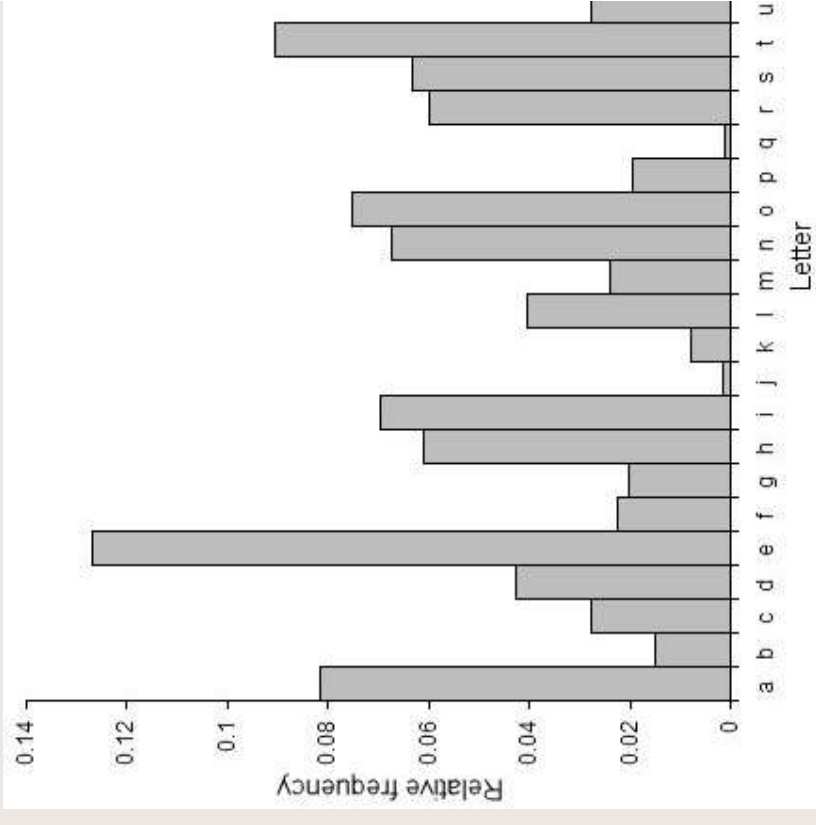
Permute column (4, 2, 1, 3) $\rightarrow$ (1, 2, 3, 4) and then row (3, 2, 1) $\rightarrow$ (1, 2, 3)

$$\begin{bmatrix} N & A & D & W \\ T & K & C & A \\ A & T & A & T \end{bmatrix} \rightarrow \begin{bmatrix} D & A & W & N \\ C & K & A & T \\ A & T & T & A \end{bmatrix} \rightarrow \begin{bmatrix} A & T & T & A \\ C & K & A & T \\ D & A & W & N \end{bmatrix}$$

Plaintext: 'attackatdawn'

Cryptanalysis of Double Transposition Cipher

- The transposition cipher is not the easiest of transposition ciphers to break, but there are statistical properties of language that can be exploited to recover the key.
- Cracking by hand is usually performed by anagramming, or trying to reconstruct the route. The more complex the route, the more difficult to crack.



One-Time Pad

The one-time pad, which is also known as the Vernam cipher.

Let P be the plaintext and K is the key.

- **Encryption** $C = E_k(P) = (P \oplus K)$
- **Decryption** $P = D_k(C) = (C \oplus K)$
- Consider an alphabet with only eight letters and the corresponding binary representation

letter	e	h	i	k	l	r	s	t
binary	000	001	010	011	100	101	110	111

- Key: randomly selected string of bits that is the same length as the message

One-Time Pad

Suppose that Alice has the key 111 101 110 101 111 100 000 101 110 000
 Encrypt her message ‘**heihitler**’.

	h	e	i	l	h	i	t	l	e	r
plaintext:	001	000	010	100	001	010	111	100	000	101
key:	111	101	110	101	111	100	000	101	110	000
ciphertext:	110	101	100	001	110	110	111	001	110	101
	s	r	l	h	s	s	t	h	s	r

- the ciphertext ‘**srlhssthsr**’ sent to Bob.

Bob decrypts using same shared key to recovers the plaintext:

	s	r	l	h	s	s	t	h	s	r
ciphertext:	110	101	100	001	110	110	111	001	110	101
key:	111	101	110	101	111	100	000	101	110	000
plaintext:	001	000	010	100	001	010	111	100	000	101
	h	e	i	l	h	i	t	l	e	r

One-Time Pad

Charlie claims that the actual key used to encrypt Alice's message is

101 111 000 101 111 100 000 101 110 000.

Bob decrypts the ciphertext using the key given to him by Charlie and obtains

	s	r	l	h	s	s	t	h	s	r
ciphertext:	110	101	100	001	110	110	111	001	110	101
“key”:	101	111	000	101	111	100	000	101	110	000
“plaintext”:	011	010	100	100	001	010	111	100	000	101
k	i	l	l	h	i	t	l	e	r	

- One-time pad is provably secure
- The key (or pad) must be chosen at random, used only once,

Cryptanalysis of One Time Pad

- Why The key (or pad) used only once?
- Two plaintext P_1 and P_2 encrypted these as $C_1 = P_1 \oplus K$ and $C_2 = P_2 \oplus K$, with the same "pad" K . This is known as a *depth*.
- $C_1 \oplus C_2 = P_1 \oplus K \oplus P_2 \oplus K = P_1 \oplus P_2$
- The key has disappeared from the problem. This not good for anyone except for Trudy.
- If two messages are in depth, for the correct key, both putative plaintexts must make sense.
- This provides the attacker with a means to distinguish the correct key from incorrect guesses. The problem only gets worse (or better, from Trudy's perspective) the more times the key is reused.

Cryptanalysis of One Time Pad

Consider an example of one-time pad encryptions that are in depth

$$P_1 = \text{like} = 100010011000 \quad \text{and} \quad P_2 = \text{kite} = 011010111000$$

both are encrypted with the same key is

$$K = 110011101111. \quad \text{Then}$$

	l	i	k	e
P_1 :	100	010	011	000
K :	110	011	101	111
C_1 :	010	001	110	111

	k	i	t	e
P_2 :	011	010	111	000
K :	110	011	101	111
C_2 :	101	001	010	111

If Trudy knows that C_1 and C_2 are in depth, she immediately sees that the second and fourth letters are the same, since the corresponding ciphertext letters are identical.

Cryptanalysis of One Time Pad

Trudy assumes a message P_1 and check her results using P_2 .

Suppose that Trudy suspects that $P_1 = \text{kill} = 011\ 010\ 100\ 100$ and find the corresponding p key:

	k	i	l	l
putative P_1 :	011	010	100	100
C_1 :	010	001	110	111
putative K :	001	011	010	011

C_2 :	101	001	010	111
putative K :	001	011	010	011
putative P_2 :	100	010	000	100
	l	i	e	l

Guess of P_1 is incorrect because decryption is not sensible.

When Trudy eventually guesses $P_1 = \text{like}$, will obtain the correct key K and decrypt to find kite, thereby confirming the correctness of the key and, therefore, the correctness of the decryption.

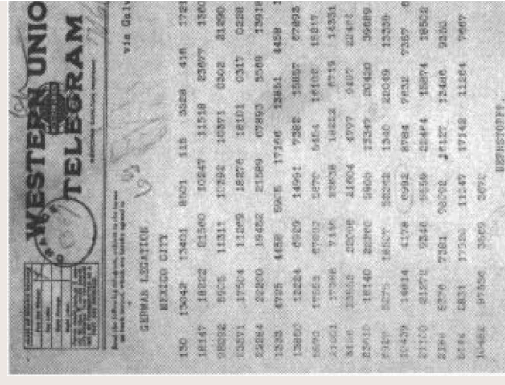
Codebook Cipher

- A classic codebook cipher is, book that is containing (plaintext) words and their corresponding (ciphertext) codewords.
- To encrypt a given word, simply look up the word in the codebook and replace it with the corresponding codeword.

- Decryption, using the inverse codebook, was equally straightforward.

- Table contains an excerpt from a famous codebook used by Germany during World War I.

Plaintext	Ciphertext
Februar	13042
fest	18147
finanzielle	98332
folgender	13303
Frieden	13800
Friedenschluss	14501
:	7382
:	1513



Codebook Cipher of the U.S. presidential Election of 1

- One of the ciphers used was a partial codebook together with a transposition on the words. The codebook was only applied to important words and the transposition was a fixed permutation for all messages of a given length.
- **Example:** The permutation used for a message of 10 words was **9,3,6,1,10,5,2,7,4,8**.
- One actual ciphertext message was

Warsaw they read all unchanged last are idiots can't situation

- Decrypted by undoing the permutation and substituting **telegram** for **Warsaw** to obtain

Election of 1876 Code

Can't read last telegram.

Situation unchanged.

They are all idiots.

Plaintext	Ciphertext
Greenbacks	Copenhagen
Hayes	Greece
votes	Rochester
Tilden	Russia
telegram	Warsaw
:	:

Cryptanalysis of Codebook cipher

A cryptanalyst could therefore compare all messages of the same length, making it relatively easy to discover the fixed permutation, even without knowledge of the partial codebook.

References

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- <http://practicalcryptography.com>
- <https://crypto.interactive-maths.com>
- <https://en.wikipedia.org>